Arduino Based Hand Gesture Controlled Wheelchair



A report submitted in fulfillment of the requirements for the course

on

ETE 2200: Electronic Project Design and Development

by

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Roll No.1904049

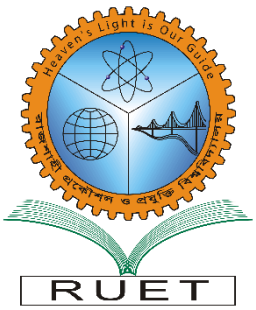
to the

Department of Electronics & Telecommunication Engineering

Faculty of Electrical & Computer Engineering

Rajshahi University of Engineering & Technology

Heaven's Light is Our Guide
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CERTIFICATE

This is to certify that the project entitled "Arduino Based Hand Gesture Controlled Wheelchair" by Saeedul Haque, Roll No. 1904049 has been carried out under my supervision. To the best of my knowledge, this project work is an original one and was not submitted anywhere for any degree or diploma.

Supervisor

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Acknowledgement

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Abstract

Gesture controlled wheelchair is based upon the concept of gesture-controlled robot. It is a type of robot which can be controlled by simple human gestures. The user just needs to wear a gesture device where a sensor is included. The sensor will record the movement of hand in a specific direction which will result in the motion of the robot in the respective directions. The robot or wheelchair is connected using wires to the gesture sensing device. The wheelchair can be controlled using accelerometer sensor. The sensor will detect the changes in X and Y axis and select the direction of the wheelchair accordingly. The sensor is intended to replace the use of external assistance or remote control. The data received from the accelerometer will be encoded and transmitted through serial communication using a wire. There will be a decoder which will decode the transmitted signal and run the motor driver according to the commands. This project is highly beneficial for the physically challenged people of our society.

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Chapter 1

Introduction

Robotics is the new booming field, which will be of great use to society in the coming years. Though robots can be a replacement to humans, they still need to be controlled by humans itself. Beyond controlling the robotic system through physical devices, recent methods of gesture control have become very popular. The main purpose of using gestures is that, it provides a more natural way of controlling and provides a rich and intuitive form of interaction with the robotic system. Human hand gestures are natural and it is easier to interact with the robot in a friendly way. This concept of gesture-controlled robot can be implemented on a wheelchair. Most wheelchair users require the assistance of other people or they have to rotate the wheels manually. But with the help of this project, a wheelchair user can simply use their hand gestures to determine the distance and direction of the wheelchair.

1.1 Objectives

For this project some objectives were taken into account in order to fulfill the requirements of physically challenged people of our society. The main objectives of this project are,

- Creating a cheap and affordable gesture-controlled wheelchair.
- Reduce the requirement of assistance for physically challenged individuals.
- Make the experience of using a wheelchair simple and comforting

Chapter 2

Background and Preliminaries

2.1 Block diagram

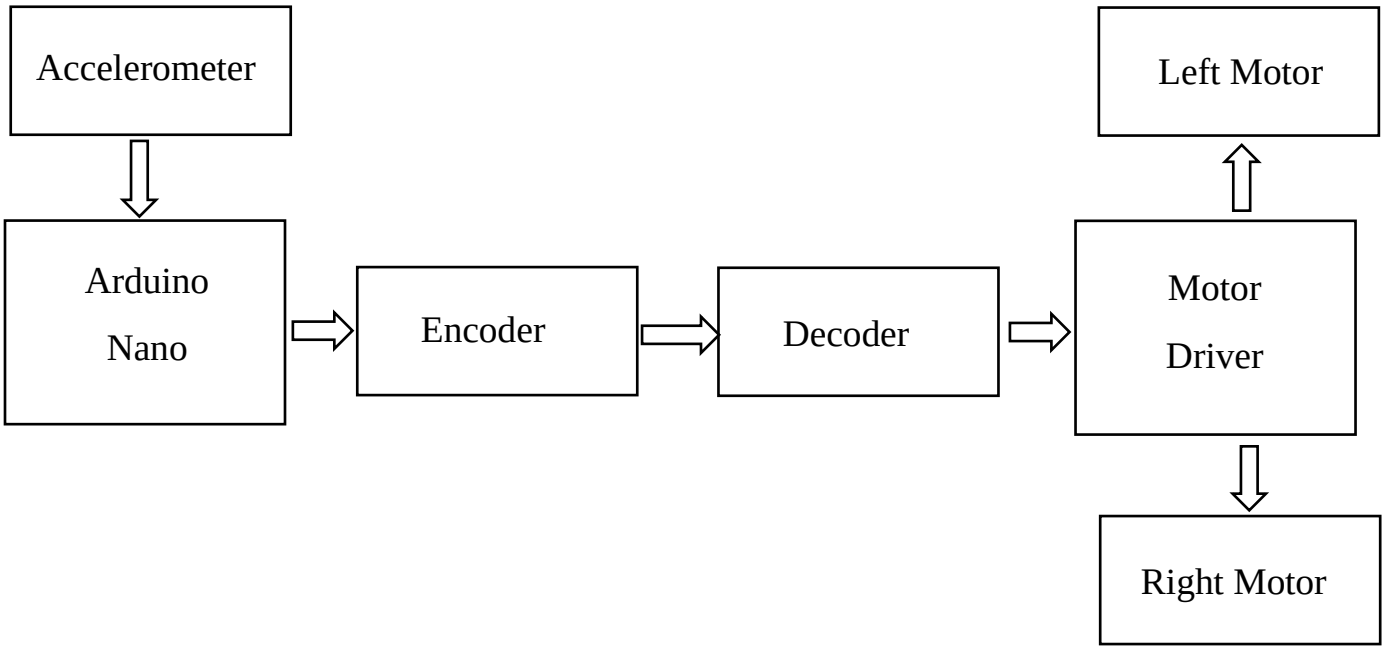


Figure 2.1: Block Diagram of the system

2.2 Component Used

- ADXL335 Analog Accelerometer
- HT12E encoder IC
- Arduino Nano
- 9V battery
- HT12D decoder IC
- L298N Motor Driver Module
- Battery operated lightweight DC Motor
- Wheel
- 3.7V lithium-ion battery
- Castor wheel
- Voltage regulator IC
- Connecting wires
- PVC board

2.3 Description of component

2.3.1 ADXL335 Analog Accelerometer

An accelerometer is a electronics component that can detect vibrations and accelerations. As we already know that acceleration is the change of velocity in a body or an object concerning time. So, an accelerometer senses these vibrations produced by that body and used this for the change of direction. It senses the acceleration and vibration and transforms them into pressure or stress; we know this phenomenon as the piezoelectric effect. After that, this energy is then get converted into an electrical voltage which is needed to drive the device. In other words, it transforms mechanical energy into electrical energy. It is a module that takes the analog input and measures the acceleration in the three dimensions, which are X, Y, and Z. Therefore, used in the construction machines like drilling machines. Also, in the human sports devices like running, walking, etc. In short, this low-power, low-noise module can easily be linked with any microcontroller or microprocessors.[1]

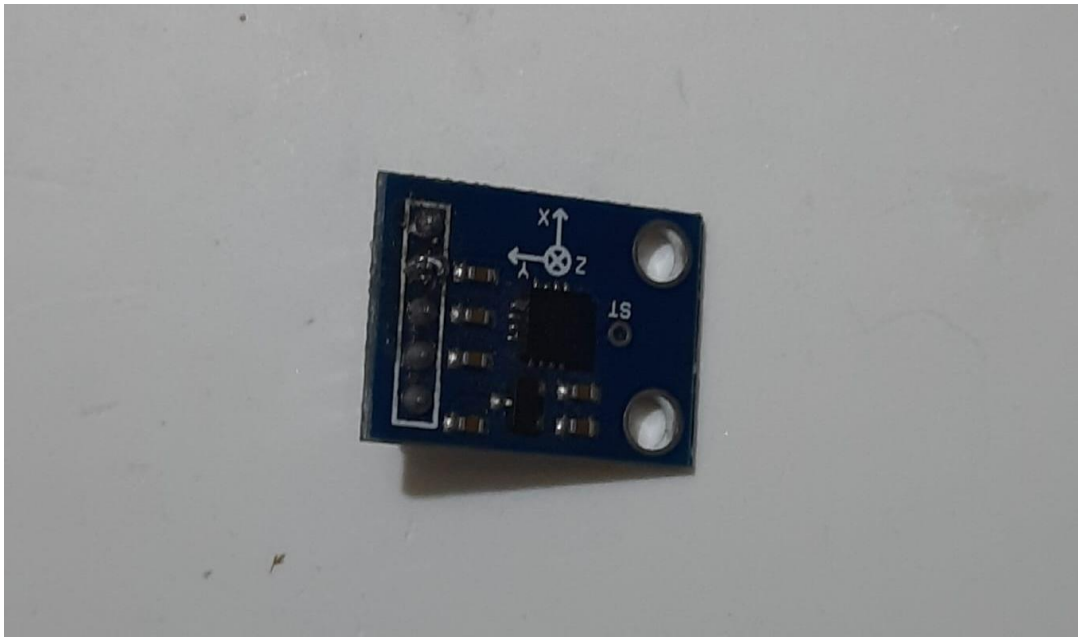


Figure 2.2: ADXL335 Analog Accelerometer

Features and Specifications:

- Operating Voltage: 3V to 6V DC
- Operating Current: 350μA
- Sensing Range: ±3g
- 3-axis sensing
- Needs no external components
- Small, cheap and easily available
- Single-supply operation: 1.8 V to 3.6 V
- 10,000 g shock survival
- Excellent temperature stability

Pin Name	Description
VCC	Pin for powering the module requires 5V
GND	Common ground pin, Ground for supply
X	X-axis analog output pin
Y	Y-axis analog output pin
Z	Z-axis analog output pin
ST	Pin that controls self-test feature

2.3.2 Arduino Nano

Arduino Nano is a type of microcontroller board, and it is designed by Arduino.cc. It can be built with a microcontroller like Atmega328. It is a small size board and also flexible with a wide variety of applications. This Nano board is different in packaging. It doesn't have any DC jack so that the power supply can be given using a small USB port otherwise straightly connected to the pins like VCC & GND. This board can be supplied with 6 to 20volts using a mini-USB port on the board.

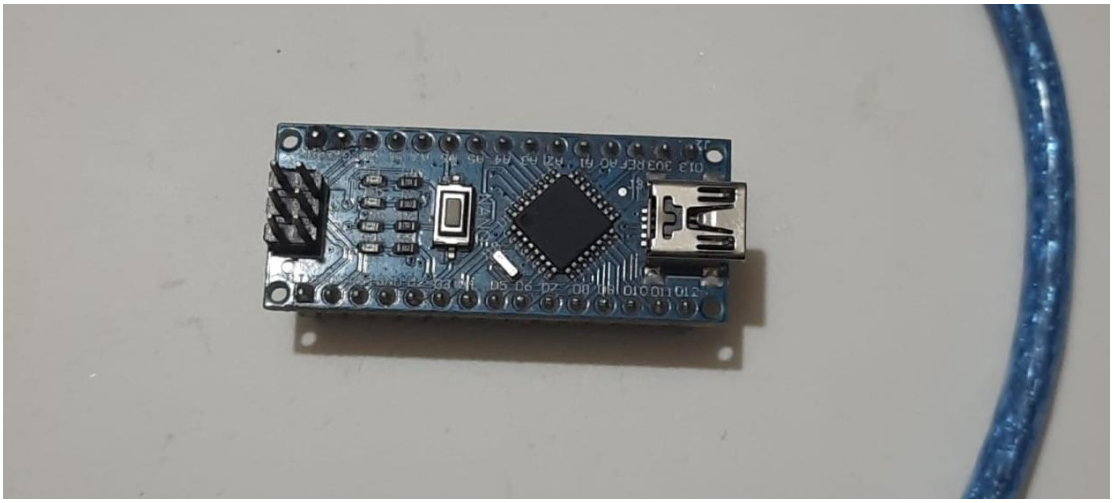


Figure 2.3: Arduino Nano

Features and specifications:

- ATmega328P Microcontroller is from 8-bit AVR family
- Operating voltage is 5V
- Input voltage (Vin) is 7V to 12V
- Analog i/p pins are 6 from A0 to A5
- Digital pins are 14
- Power consumption is 19 mA
- I/O pins DC Current is 40 mA
- Flash memory is 32 KB
- SRAM is 2 KB
- EEPROM is 1 KB
- CLK speed is 16 MHz
- Weight-7g
- Supports three communications like SPI, IIC, & USART

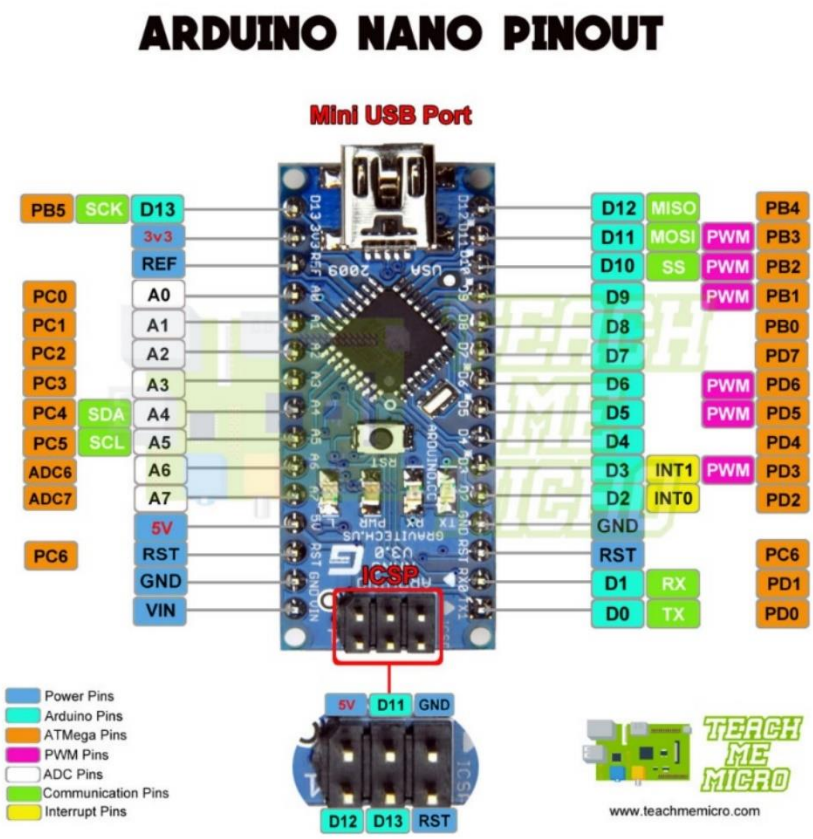


Figure 2.4: Arduino Nano Pinout [2]

2.3.3 HT12E Encoder IC

HT12E is an encoder integrated circuit of 2^{12} series of encoders. They are paired with 2^{12} series of decoders for use in remote control system applications. It is mainly used in interfacing RF and infrared circuits. The chosen pair of encoder/decoder should have same number of addresses and data format. Simply put, HT12E converts the parallel inputs into serial output. It encodes the 12-bit parallel data into serial for transmission. These 12 bits are divided into 8 address bits and 4 data bits. HT12E has a transmission enable pin which is active low. When a trigger signal is received on TE pin, the programmed addresses/data are transmitted together with the header bits via an RF or an infrared transmission medium. HT12E begins a 4-word transmission cycle upon receipt of a transmission enable. This cycle is repeated as long as TE is kept low. As soon as TE returns to high, the encoder output completes its final cycle and then stops.[3]

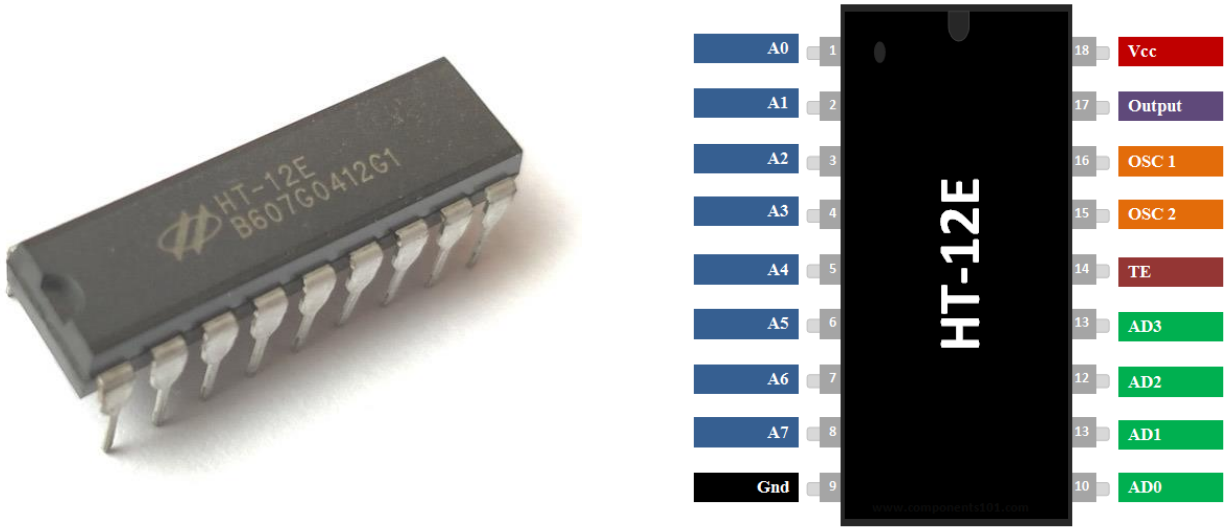


Figure 2.5: HT12E IC Pinout [4]

Pin Number	Pin Name	Description
1 to 8	A0,A1,A2,A3,A4,A5,A6 & A7	These are the 8-bit address bits, which is used to protect your data. We should set the bits in same pattern on Encoder and Decoder IC to pair them.
9	Ground/Vss	Connected to the Ground of circuit
10 to 13	AD0, AD1, AD2 & AD3	These four pins are used to send data, the data encoded here will be decoded on HT12D IC sharing the same address bits
14	Transmission Enable(TE)	This pin has to be connected to Ground (0V) to enable the Transmission.
15 and 16	Oscillator pins 1 & 2	The IC has a built-in oscillator. This oscillator can be used by connecting these two pins through a 1M Resistor

17	Output	The Encoded 12 bit output data can be obtained from this pin
16	Vcc/Vdd	This pin powers the IC, typically +5V is used. Can range from 2.4V to 12V

2.3.4 HT12D Decoder IC

HT12D is a decoder integrated circuit that belongs to 2^{12} series of decoders. It is mainly provided to interface RF and infrared circuits. They are paired with 2^{12} series of encoders. The chosen pair of encoder/decoder should have same number of addresses and data format. In simple terms, HT12D converts the serial input into parallel outputs. It decodes the serial addresses and data received by a receiver, into parallel data and sends them to output data pins. The serial input data is compared with the local addresses three times continuously. The input data code is decoded when no error or unmatched codes are found. A valid transmission in indicated by a high signal at VT pin. HT12D is capable of decoding 12 bits, of which 8 are address bits and 4 are data bits. The data on 4-bit latch type output pins remain unchanged until new is received.[5]

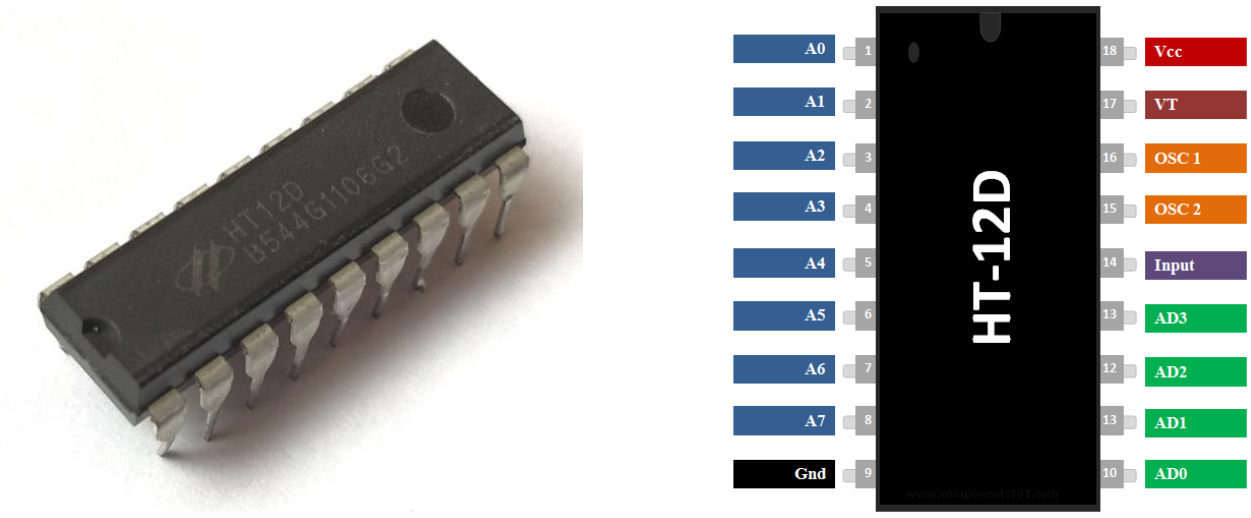


Figure 2.6: HT12D IC Pinout[6]

Pin Number	Pin Name	Description
1 to 8	A0,A1,A2,A3,A4,A5,A6 & A7	These are the 8-bit address bits, which is used to protect your data. We should set the bits in same pattern on Encoder and Decoder IC to pair them.
9	Ground/Vss	Connected to the Ground of circuit
10 to 13	AD0, AD1, AD2 & AD3	These four pins are used to obtain the data bits by decoding the data from HT12E IC
14	Input	The Encoded 12 bit output data obtained from HT12E has to be given here
15 and 16	Oscillator pins 1 & 2	The IC has a built in oscillator. This oscillator can be used by connecting these two pins through a 1M Resistor
17	Valid Transmission (VT)	This pin will go high when a data is received. It is not mandatory to use it.
16	Vcc/Vdd	This pin powers the IC should use only 5V

2.3.5 L298N Motor Driver Module

L298N module is a high voltage, high current dual full-bridge motor driver module for controlling DC motor and stepper motor. It can control both the speed and rotation direction of two DC motors. This module consists of an L298 dual-channel H-Bridge motor driver IC. This module uses two techniques for the control speed and rotation direction of the DC motors. These are PWM – For controlling the speed

and H-Bridge – For controlling rotation direction. These modules can control two DC motor or one stepper motor at the same time.

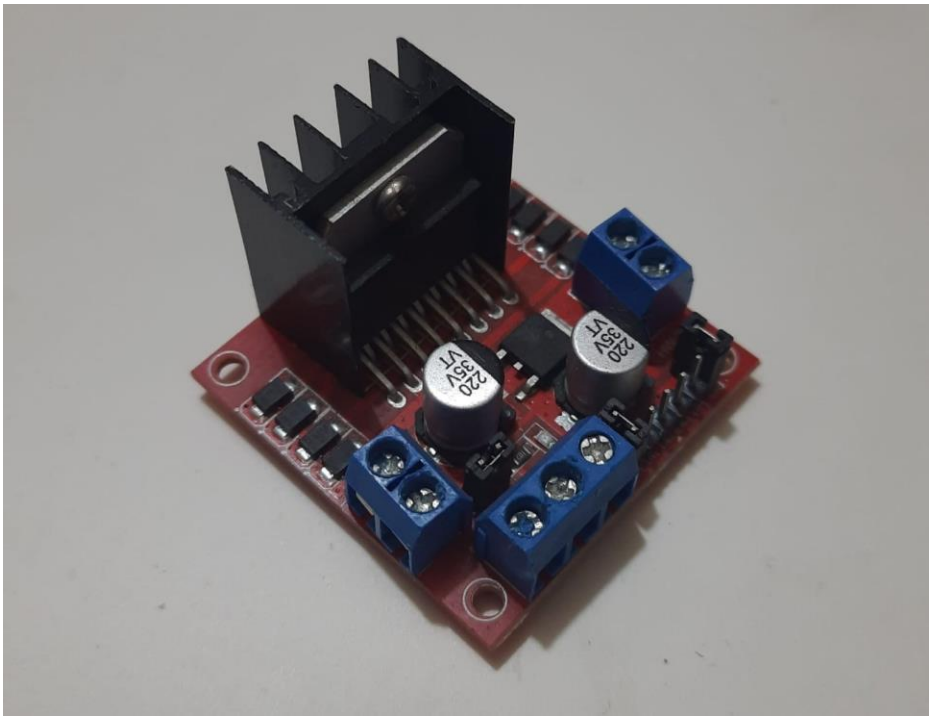


Figure 2.7: L298N Motor Driver Module

Pin Name	Description
IN1 & IN2	Motor A input pins. Used to control the spinning direction of Motor A
IN3 & IN4	Motor B input pins. Used to control the spinning direction of Motor B
ENA	Enables PWM signal for Motor A
ENB	Enables PWM signal for Motor B
OUT1 & OUT2	Output pins of Motor A
OUT3 & OUT4	Output pins of Motor B
12V	12V input from DC power Source
5V	Supplies power for the switching logic circuitry inside L298N IC
GND	Ground pin

2.3.6 Battery operated DC Motor

A DC motor is an electric motor that runs on direct current power. In an electric motor, the operation is dependent upon simple electromagnetism. A current-carrying conductor generates a magnetic field, when this is then placed in an external magnetic field, it will encounter a force proportional to the current in the conductor and to the strength of the external magnetic field. It is a device that converts electrical energy to mechanical energy. It works on the fact that a current-carrying conductor placed in a magnetic field experiences a force that causes it to rotate with respect to its original position. Practical DC Motor consists of field windings to provide the magnetic flux and armature which acts as the conductor. The operating voltage range of this type of motor can range from 3 to 12 volts. It can produce a speed of up to 200 rpm.

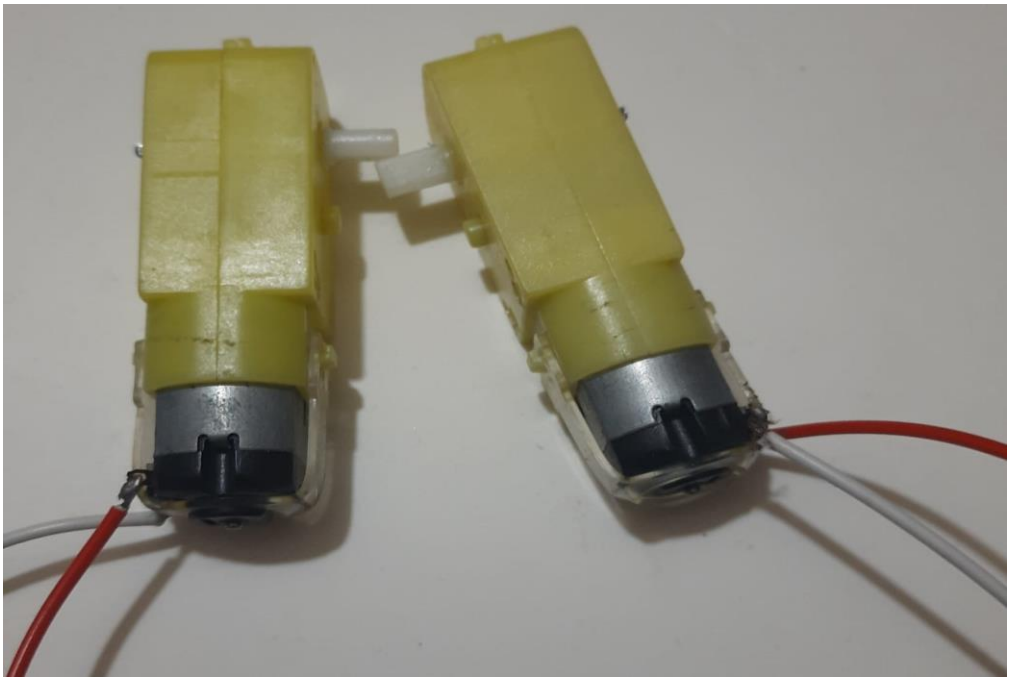


Figure 2.8: Battery operated DC Motor

2.3.7 Lithium-ion battery (3.7V)

A lithium-ion or Li-ion battery is a type of rechargeable battery which uses the reversible reduction of lithium ions to store energy. A 3.7-volt lithium-ion battery can last anywhere from two to six hours, depending on the power of the battery and the device it is powering. A higher capacity battery will last longer than a lower

capacity one, and a device that uses more power will drain the battery faster than one that uses less power



Figure 2.9: Lithium-ion battery (3.7V)

2.4 Software Used

- Arduino IDE
- ExpressPCB

2.5 Description of software

2.5.1 Arduino IDE

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++. Here, IDE stands for Integrated Development Environment. The program or code written in the Arduino IDE is often called as sketching. We need to connect the Genuino and Arduino board with the IDE to upload the sketch written in the Arduino IDE software. The sketch is saved with the extension '.ino.'

2.5.2 ExpressPCB

ExpressPCB is CAD software that is used to create layouts for printed circuit boards (PCB). The CAD is divided into the more helpful category such as ExpressSCH and ExpressPCB. ExpressSCH is used to design the schematics. ExpressPCB to create a PCB design. It is linked to the schematic.

Chapter 3

Project Details

3.1 Working Principle

The gesture-controlled wheelchair has an analog accelerometer (ADXL335) which senses the change in different axes and transmits this data to Arduino Nano. This data is then encoded using the encoder. Hence the parallel data is converted into a single serial output. The Arduino is powered a single 9V battery. The Arduino then powers the other components such as the encoder IC and accelerometer. The encoded data is sent using a wire to the decoder which is connected to the motor driver. When the data reaches the decoder, it is decoded and applied to the motor driver. According to the data received, the left and right motor will rotate accordingly. The motor driver was powered using two 3.7V battery connected in series or 7.4V of DC voltage.

The movement of the wheelchair is as follows-

- The wheelchair will move forward when the X axis value is greater than 390
- The wheelchair will move backward when the X axis value is less than 310
- The wheelchair will move to the left when Y axis value is less than 320
- The wheelchair will move to the right when Y axis value is greater than 380

The required value for changing directions can be changed according to the user's requirements.

3.2 Circuit Diagram

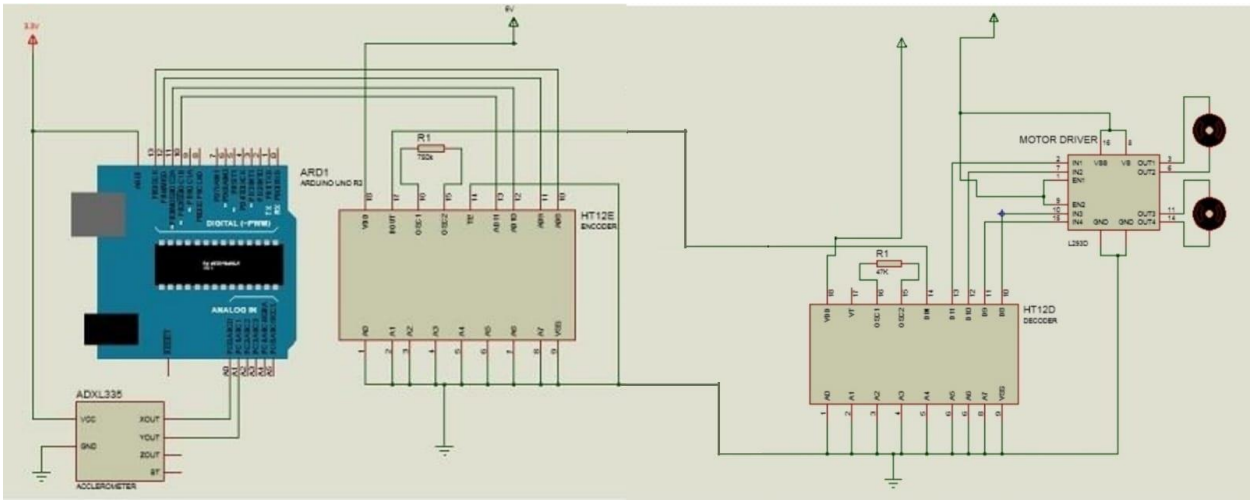


Figure 3.1: Circuit diagram

3.3 PCB design

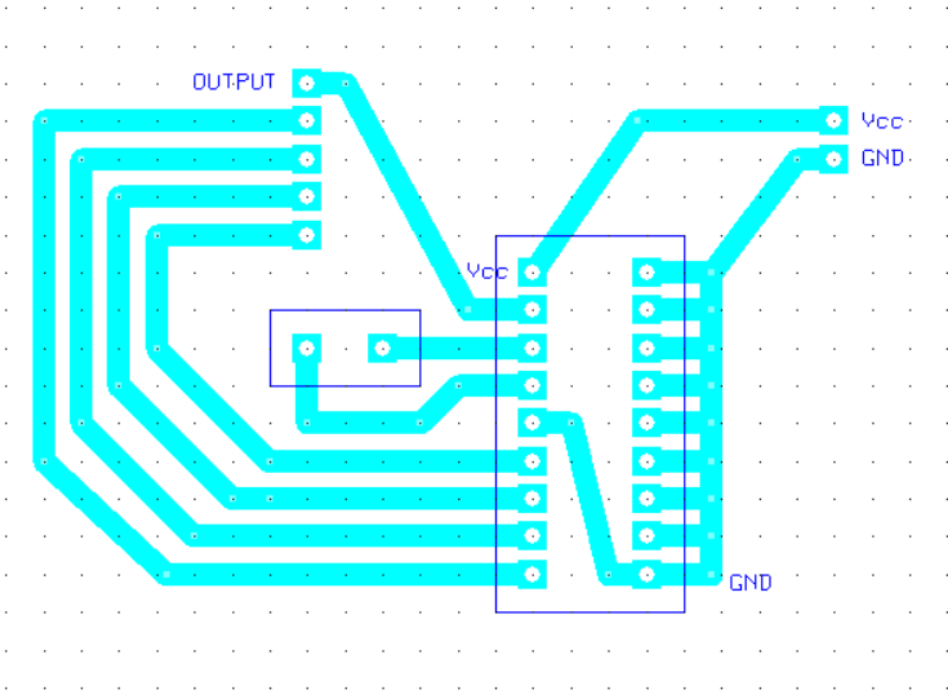


Figure 3.2: PCB design of encoder IC

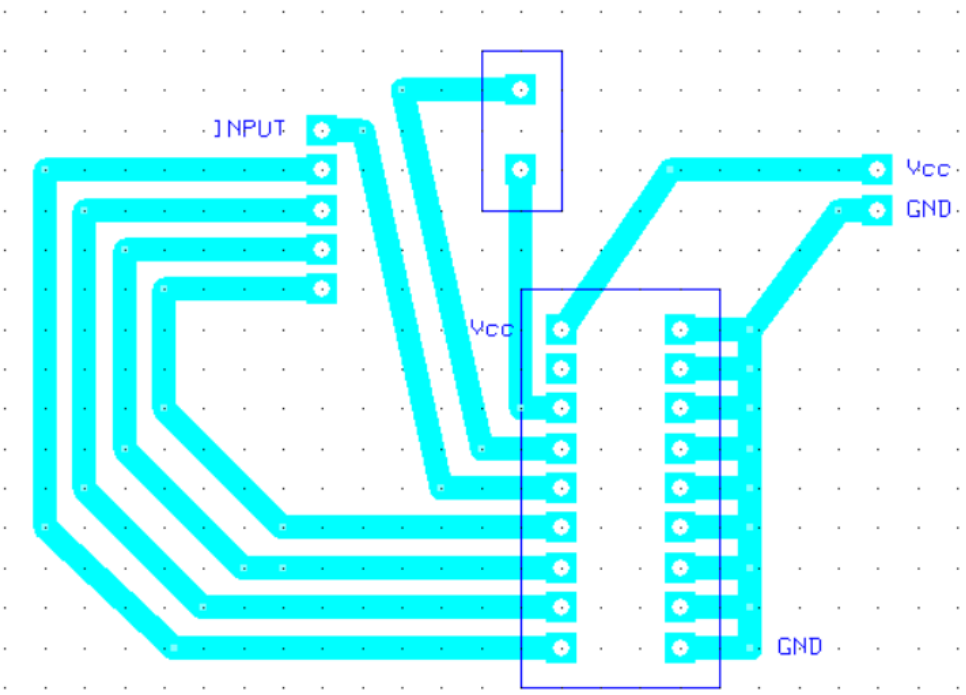
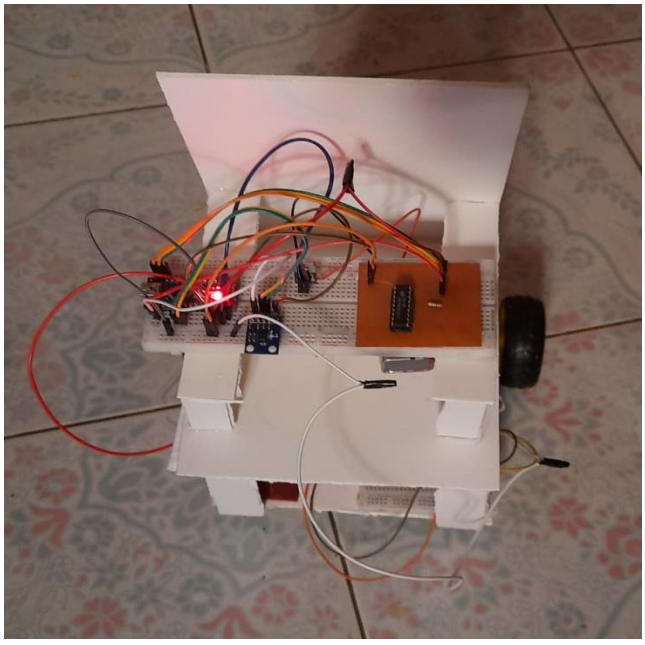


Figure 3.3: PCB design of decoder IC

3.4 Result



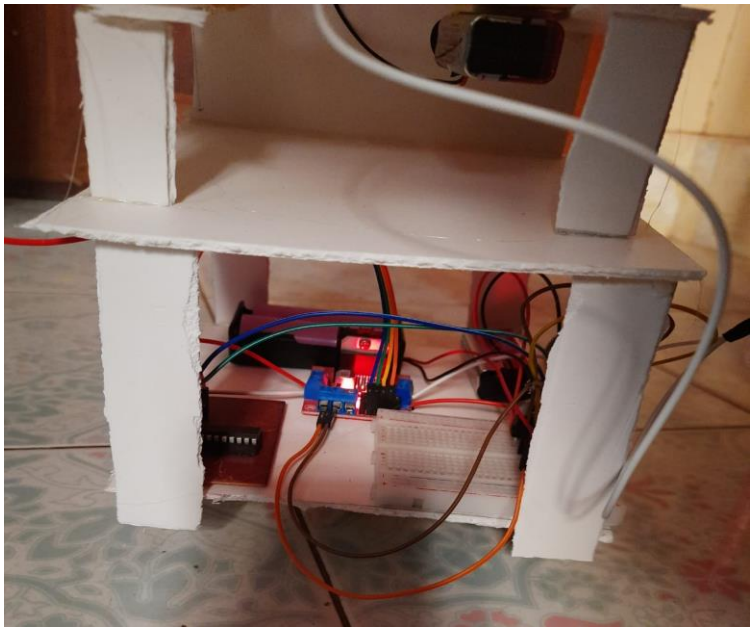


Figure 3.4: Prototype model of the project

Chapter 4

Conclusion

4.1 Advantage

- Easy to operate
- User friendly
- Low power consumption
- Components easily available
- Low cost

4.2 Disadvantage

- Power supply failure will result in system shutdown
- If any component is damaged, it may cause dire consequences
- Since the system is wired, the operation range is limited

4.3 Application

- Assist physically challenged people
- General purpose device for better living

4.4 Future plan

This project has numerous scopes for upgrade in the future. It can be equipped with better motors and wheel to achieve more control over movememnt. Another way of improvement is by adding another accelerometer to the chassis to detect fall or accident. This will drastically improve the safety of the wheelchair. It can also be equipped with a health monitoring system to keep the health of the user in check. Any slight degradation of health of the user can be quickly sent to a proffessional. Also, the speed control of the wheelchair can be improved by adding another arduino to the chassis of the wheelchair to control the motor driver.

4.5 Conclusion

The main purpose of this project is to offer comfort and bring ease to the physically challenged people of our society. Here, the use of accelerometer to control the direction and movement of the wheelchair has significantly reduced the pressure and work for the users of wheelchair. Due to this project, a physically challenged person is able to easily move freely from one place to another and without the need of anyone else. The overall complexity of the project was simple. The cost of the whole project is manageable for most of the people. So, further use of this project on a large scale will definitely help rise the living quality of people.

References

[1] https://www.circuits-diy.com/adxl335-3-axis-accelerometer-module/

[2] https://www.teachmemicro.com/arduino-nano-pinout-diagram/

[3] https://www.engineersgarage.com/ht12e-encoder-ic/

[4] https://components101.com/ics/ht12e-encoder-pin-diagram-datasheet

[5] https://www.engineersgarage.com/ht12d-decoder-ic/

[6] https://components101.com/ics/ht12d-rf-decoder-ic

[7] https://github.com/bunny98/Gesture-Controlled-Robot

[8] Gomathy, C K. (2021). THE GESTURE CONTROLLED ROBOT.

Appendix

Arduino Code

```
int xPin=A3;
int yPin=A2;
int Q1=10,Q2=11,Q3=12,Q4=13;
long x;
long y;
long z;
void setup()
{
  Serial.begin(9600);
  pinMode(Q1,OUTPUT);
  pinMode(Q2,OUTPUT);
  pinMode(Q3,OUTPUT);
  pinMode(Q4,OUTPUT);
}
void loop()
{
  x = analogRead(xPin);
  y = analogRead(yPin);
```

```
   Serial.print(analogRead(xPin));
  Serial.print("\t");
  Serial.print(analogRead(yPin));
  Serial.print("\t");
    if(x<320){
      left();}
    else if(x>370){
      right();}
    else if(y>400){
      forward();}
    else if(y<310){
      backward();}
    else
      {stop_();}
}
void stop_()
{
  Serial.println("");
  Serial.println("STOP");
  digitalWrite(Q1,LOW);
  digitalWrite(Q2,LOW);
  digitalWrite(Q3,LOW);
  digitalWrite(Q4,LOW);
}
void forward()
{
  Serial.println("");
  Serial.println("Forward");
  digitalWrite(Q1,HIGH);
  digitalWrite(Q2,LOW);
  digitalWrite(Q3,LOW);
  digitalWrite(Q4,HIGH);
}
void backward()
{
  Serial.println("");
  Serial.println("Backward");
  digitalWrite(Q1,LOW);
  digitalWrite(Q2,HIGH);
  digitalWrite(Q3,HIGH);
```

```
  digitalWrite(Q4,LOW);
}
void left()
{
  Serial.println("");
  Serial.println("Left");
  digitalWrite(Q1,HIGH);
  digitalWrite(Q2,LOW);
  digitalWrite(Q3,HIGH);
  digitalWrite(Q4,LOW);
}
void right()
{
  Serial.println("");
  Serial.println("Right");
  digitalWrite(Q1,LOW);
  digitalWrite(Q2,HIGH);
  digitalWrite(Q3,LOW);
  digitalWrite(Q4,HIGH);
}
```